

ECON 101—Introduction to Microeconomics
Fall 2024

Midterm I

September 19, 2024

Part I: Multiple Choice

Please record all answers in the front sheet. Each question is worth 2 points for a total of 12 points.

Version	I	II	III	IV
Question				
1	(d)	(b)	(c)	(c)
2	(a)	(d)	(b)	(a)
3	(b)	(a)	(d)	(b)
4	(a)	(a)	(d)	(c)
5	(a)	(b)	(c)	(b)
6	(b)	(b)	(c)	(a)

Part II: Long Problems

Please answer *only* in the answers booklet. Be neat and give complete arguments. Points for each item are indicated. You can use as much scratch paper as you need but your scratch work will not be considered.

The whimsical land of Ekonomika

In the land of **Ekonomika**, the fair, wise, and highly knowledgeable in Economics, Princess Medina rules over her subjects. The law in this land is maintained, harshly but fairly, by Captain Akmaral and General Nazima (i.e., cheating is illegal and any suspected cheating is dealt with the harshest penalties).

In Ekonomika, unicorns are not just mythical creatures—they are hard working entrepreneurs who produce *Sparkle Dust*, which is a crucial ingredient for wizard's potions and the lattes that the Gen Z and Gen α citizens of Ekonomika like so much, as well as having many other uses. Suppose that the market for Sparkle Dust can be modeled using the tools that you have learned in ECON101.

Lord Treasurer Bagdat claims that the market for Sparkle Dust can be modeled according to the following supply and demand equations:

$$\begin{aligned}Q_d^0 &= 240 - 2P \\ Q_s^0 &= 4P - 80\end{aligned}$$

Recently, the unicorns discovered that listening to Taylor Swift songs while working doubles their Sparkle Dust production. However, constant exposure to Taylor Swift songs causes the unicorns to lose their magical rainbow colors, turning them plaid instead. This has two effects:

- (Event 1) Listening to Taylor Swift doubles unicorn productivity:

$$Q_s^1 = 8P - 160.$$

- (Event 2) Gen α consumers now prefer *Plaid Unicorn Sparkle Dust*, believing it to be more authentic, i.e., they claim that it is *mad Sigma*. This shifts the demand for Sparkle Dust:

$$Q_d^1 = 320 - 2P.$$

However, there is a problem! The Council of Wizards, led by Archmage Nuray, is concerned that the overproduction of Sparkle Dust will have deleterious effects on the space-time continuum. (Event 3) Archmage Nuray recommends to put a *per-unit tax* of \$10 on Sparkle Dust to curb its production.

(Event 4) Additionally, the evil wizard Sultan spreads a rumor that consuming Sparkle Dust grants immortality, which further affects the demand to:

$$Q_d^2 = 400 - 2P.$$

After all this craziness, Lord Treasurer Bagdat gathers two experts on the matter, Minister Abuzer and Exchequier Rakhim, to analyze this mess. In this problem, you will help them understand what is going on in the market for Sparkle Dust in the land of Ekonomika! In each task below assumes that the events prior to the one being considered remain.

1. (5 points) Calculate the **initial equilibrium** before any changes occur.
2. (5 points) Determine the new equilibrium price and quantity after the productivity boost (Event 1).
3. (5 points) Find the equilibrium price and quantity after the Gen α demand increase (Event 2).
4. (5 points) Calculate the equilibrium price and quantity after the per-unit tax is imposed (Event 3).
5. (5 points) Calculate the final equilibrium price and quantity after the immortality rumor affects demand (Event 4.)

6. (10 points) Compute consumer surplus, producer surplus, and the dead-weight loss, if any, at each stage.
7. (8 points) Discuss the welfare implications for the inhabitants of Ekonomika. In your argument, you must use the concepts of tax incidence, and market efficiency.

Note: Show all your calculations and illustrate each stage with supply and demand graphs.

The time-traveling labor market of South Park

In the quaint town of South Park, the demand for snow-shoveling services is at an all-time high during winter. The demand for labor is fixed at $Q = 100$ jobs per day, regardless of the wage rate.

Initially, only the local workers, affectionately known as **Townies**, supply labor in this market. Their supply curve is given by:

$$S_{\text{Townies}} : W = 0.5Q_T + 5$$

where: W is the wage rate in dollars per hour, and Q_T is the quantity of labor supplied by the Townies.

One frosty morning, a group of time-traveling workers from the future, dubbed **Future Folks**, arrive and begin offering their labor. Due to difficulties in obtaining formal snow-shoveling jobs, they can supply a maximum of $Q_F^{\max} = 30$ jobs per day. Their supply curve is given by

$$S_{\text{Future Folks}} : W = 0.3Q_F + 2 \quad \text{with } Q_F \leq 30.$$

where: Q_F is the quantity of labor supplied by the Future Folks.

The Townies are up in arms, exclaiming, "They took our jobs!"

To address the uproar, the South Park town council decides to implement a minimum wage policy set at $W_{\min} = \$50$ per hour.

After this policy is enacted, the townies are still up in arms claiming: **"They took our jobs!"**



1. (20 points) **Without** the minimum wage, determine the equilibrium wage rate and the total quantity of labor supplied. How many jobs do the Townies and Future Folks each supply at this wage?
2. (25 points) **With** the minimum wage of \$50 per hour in place, analyze its effect on the labor market. Consider the following allocation methods:
 - (a) Method 1: Assuming no hiring bias, jobs could be allocated proportionally on a first-come, first-served basis. This means that given the total labor supply at the minimum wage, you will compute the proportion of townies and future-folk that want to offer their labor. You will use this proportion to compute how many townies and future-folk get the 100 available jobs.
 - (b) Method 2: Given the local uproar, employers prioritize Townies.

Discuss, under each method, whether the Townies' complaint that "They took our jobs!" is justified. Does the minimum wage help or hurt their employment prospects?

Note: Show all your calculations and illustrate each stage with supply and demand graphs.